

The Professional Programmer

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WHAT IS A PROFESSIONAL PROGRAMMER?

The single most important trait of a professional programmer is *personal responsibility*. Professional programmers take responsibility for their career, their estimates, their schedule commitments, their mistakes, and their workmanship. A professional programmer does not pass that responsibility off on others.

- *If you are a professional, then you are responsible for your own career.* You are responsible for reading and learning. You are responsible for staying up to date with the industry and the technology. Too many programmers feel that it is their employer's job to train them. Sorry, this is just dead wrong. Do you think doctors behave that way? Do you think lawyers behave that way? No, they train themselves on their own time, and their own nickel. They spend much of their off-hours reading journals and decisions. They keep themselves up to date. And so must we. The relationship between you and your employer is spelled out nicely in your employment contract. In short: your employer promises to pay you, and you promise to do a good job.
- *Professionals take responsibility for the code they write.* They do not release code unless they *know* it works. Think about that for a minute. How can you possibly consider yourself a professional if you are willing to release code that you are not sure of? Professional programmers expect QA to find *nothing* because *they don't release their code until they've thoroughly tested it*. Of course, QA will find some problems, because no one is perfect. But as professionals, our *attitude* must be that we will leave nothing for QA to find.



- *Professionals are team players.* They take responsibility for the output of the whole team, not just their own work. They help one another, teach one another, learn from one another, and even cover for one another when necessary. When one teammate falls down, the others step in, knowing that one day they'll be the ones to need cover.
- *Professionals do not tolerate big bug lists.* A huge bug list is sloppy. Systems with thousands of issues in the issue-tracking database are tragedies of carelessness. Indeed, in most projects, the very need for an issue-tracking system is a symptom of carelessness. Only the very biggest systems should have bug lists so long that automation is required to manage them.
- *Professionals do not make a mess.* They take pride in their workmanship. They keep their code clean, well structured, and easy to read. They follow agreed-upon standards and best practices. They never, *ever* rush. Imagine that you are having an out-of-body experience watching a doctor perform open-heart surgery on *you*. This doctor has a *deadline* (in the literal sense). He must finish before the heart-lung bypass machine damages too many of your blood cells. How do you want him to behave? Do you want him to behave like the typical software developer, rushing and making a mess? Do you want him to say, "I'll go back and fix this later"? Or do you want him to hold carefully to his disciplines, taking his time, confident that his approach is the best approach he can reasonably take. Do you want a mess, or professionalism?

Professionals are responsible. They take responsibility for their own careers. They take responsibility for making sure their code works properly. They take responsibility for the quality of their workmanship. They do not abandon their principles when deadlines loom. Indeed, when the pressure mounts, professionals hold ever tighter to the disciplines they know are right.